Extending BGM for Skew, Smile, and Negative Rates

15.1 What the pure lognormal model gets wrong

Chapter 11 built BGM on a pure lognormal postulate specifically because it reproduces Black's caplet formula exactly. But Chapter 8 already showed that the market smile is not flat: a single constant σ_i per rate implies a flat implied volatility across strike, which real cap and swaption markets do not exhibit, and cannot handle a rate at or below zero, which real markets have exhibited for extended periods. Two extensions, used in combination in current practice, fix these problems without discarding the drift-derivation machinery of Chapter 13.

15.2 Displaced diffusion: fixing the zero boundary and adding skew

The simplest fix replaces the lognormal SDE with a **displaced (or shifted) diffusion**.

Key result: the displaced-diffusion BGM rate

$$\mathrm{d}L_i(t) = \sigma_i(t)\,\big(L_i(t) + c_i\big)\,\mathrm{d}W_i^{i+1}(t)$$

for a per-rate **displacement** $c_i \geq 0$, under L_i's own forward measure. The rate itself, $L_i(t)$, can now go as low as $-c_i$ before the model breaks down — pushing the structural floor of Chapter 1's "lognormal can't go negative" problem down by the displacement, rather than eliminating it. All the drift-derivation machinery of Chapter 13 carries through essentially unchanged, with L_j replaced by $L_j + c_j$ throughout the drift formula.

Displaced diffusion sits on a spectrum with two named endpoints: $c_i \to \infty$ (with σ_i rescaled appropriately) recovers pure normal (Bachelier) dynamics, while $c_i = 0$ recovers the pure lognormal model of Chapter 11. A finite, positive c_i interpolates between the two, and — as a direct consequence — generates *skew*: because the effective volatility of L_i itself, $\sigma_i(L_i + c_i)/L_i$, changes as L_i moves, the implied Black volatility a displaced-diffusion model produces is no longer flat across strike, but slopes in a direction and magnitude controlled by the size of c_i relative to L_i.

Worked example 15.1 — choosing a displacement.
A desk wants a BGM model on a rate currently at $L_i(0) = 3.80\%$ that can plausibly go as low as -1.00% without breaking, matching a stress scenario the risk team has specified. What displacement c_i is needed, and what does this imply for the model's effective lognormal volatility today, given a target normal-equivalent vol of $\sigma_n = 85\,\text{bp/yr}$?
Step 1 — set the floor. The model floor is $-c_i$, so $-c_i \leq -1.00\%$ requires $c_i \geq 1.00\%$. Take $c_i = 1.00\%$ exactly.
Step 2 — convert the target normal vol to the displaced-lognormal σ_i. Near the current level, $\sigma_n \approx \sigma_i\,(L_i + c_i)$ (the same style of approximation as Chapter 1's normal/lognormal

conversion, now applied to the *shifted* rate $L_i + c_i$ rather than L_i itself):

$$\sigma_i \approx \frac{\sigma_n}{L_i + c_i} = \frac{0.0085}{0.0380 + 0.0100} = \frac{0.0085}{0.0480} = 0.1771 \;\rightarrow\; 17.7\%$$

Sanity check. Compare to the undisplaced conversion from Chapter 1: $\sigma_n/L_i = 0.0085/0.0380 = 22.4\%$. The displaced version gives a lower effective σ_i, which makes sense — the same absolute (bp) uncertainty is now being expressed as a percentage of a *larger* base ($L_i + c_i$ rather than L_i alone), so the required percentage volatility is smaller. *Interpretation.* With $c_i = 1.00\%$ and $\sigma_i = 17.7\%$, this rate can mathematically reach -1.00% (matching the risk team's stress floor) while still pricing today's at-the-money caplet at the correct 85 bp normal vol. Choosing c_i is therefore not a purely statistical fitting exercise — it is also a governance decision about what range of outcomes the model must remain well-defined for.

PITFALL

Displacement is sometimes treated as a purely cosmetic fix — "just add a shift so the log doesn't blow up" — with c_i chosen arbitrarily large for safety. This is a mistake: because c_i directly controls the skew the model produces (a larger displacement pushes the model's implied skew closer to the symmetric, skew-free Bachelier limit), an arbitrarily large displacement silently discards the model's ability to match a genuinely skewed market smile, even though every ATM price still looks fine. A displacement chosen for numerical safety alone, without checking its effect on the fitted smile shape away from the money, will underperform specifically on the instruments — low-strike floors, deep-out-of-the-money payers — where getting the skew right matters most.

15.3 Stochastic volatility: fixing the symmetric wings

Displacement alone generates skew but not the symmetric wing-widening (fatter tails on both sides) that Chapter 8 attributed to genuine stochastic volatility. The natural fix couples BGM to a SABR-style volatility process directly.

Key result: SABR–LMM (sketch)

$$\mathrm{d}L_i(t) = \sigma_i(t)\,\alpha(t)\,\big(L_i(t) + c_i\big)\,\mathrm{d}W_i^{i+1}(t), \qquad \mathrm{d}\alpha(t) = \nu\,\alpha(t)\,\mathrm{d}Z(t)$$

with $\mathrm{d}W_i\,\mathrm{d}Z = \rho_i\,\mathrm{d}t$, sharing a single common stochastic volatility multiplier $\alpha(t)$ (or, in richer variants, one per rate) across the curve. This nests the pure BGM model of Chapter 11 exactly at $\nu = 0$ (deterministic $\alpha \equiv 1$), and it independently controls skew (via ρ_i, layered on top of whatever skew the displacement c_i already contributes) and symmetric wing width (via ν), giving the model the same qualitative smile-shaping levers that SABR itself offers for a single rate, now applied jointly and consistently across the whole curve.

ON THE DESK

SABR–LMM and its close relatives are genuinely used in production at sophisticated rates desks, but they come at a real cost: the drift derivation of Chapter 13 must now also account for the stochastic volatility factor's own dynamics, closed-form caplet pricing is lost (an asymptotic expansion, in the spirit of Hagan's SABR formula, is typically used instead), and calibration — already a substantial undertaking for plain BGM, as Part V will show — gains an entire additional dimension of free parameters. Many desks make a deliberate, pragmatic choice to run plain displaced-diffusion BGM for the bulk of their book, calibrated carefully to ATM and near-ATM levels, and reserve the added complexity of a stochastic-volatility extension specifically for products whose payoff is genuinely, materially sensitive to the wings of the smile.

15.4 What the recalibration application actually needs

For the purposes of Part V, it is worth being explicit about scope: the calibration machinery built there — caplet stripping, swaption matching, cascade algorithms — applies with only minor modification to either the plain lognormal model of Chapter 11, the displaced-diffusion extension of this chapter, or (with materially more work) a full SABR–LMM. The book builds the calibration story primarily around displaced-diffusion BGM, both because it is the more common production default and because its calibration problem already contains, in miniature, every genuine difficulty a stochastic-volatility extension adds, without the additional dimension of complexity obscuring the core ideas.

Chapter benchmark

Before moving on, you should be able to, without notes:

1. Write the displaced-diffusion SDE for a BGM forward rate and explain what the displacement c_i does to the model's negative-rate floor.
2. Explain why displacement alone generates skew but not symmetric wing-widening, and why a stochastic-volatility extension like SABR–LMM is needed for the latter.
3. State, in one sentence, the practical trade-off a desk faces between plain displaced-diffusion BGM and a full stochastic-volatility extension.